

Logistic Report for Esports Tournament Network Infrastructure at UTeM

1. Project Overview

This project proposes the organization of an Esports Tournament at Universiti Teknikal Malaysia Melaka (UTeM). The event is designed to accommodate a minimum of 200 participants, including players, administrators, and spectators. The objective is to provide a stable, high-performance, and secure network environment that supports competitive gaming, event management, live streaming, and audience connectivity.



2. Venue Selection

Proposed Venue

Dewan Canselor, Universiti Teknikal Malaysia Melaka (UTeM)

Justification

Dewan Canselor was selected because it is one of the largest facilities within the university and is regularly used for large-scale academic and public events. The venue provides:

- * Large open space for gaming stations and audience seating.
- * Adequate electrical infrastructure.
- * Centralized air-conditioning system.
- * Multiple access points for network deployment.

* Easy accessibility for participants and visitors.



3. User Distribution

The tournament is designed to support a minimum of 200 users.

User Category	Number of Users
Players	50
Administrators	20
Spectators	130
Total	200



4. Network Requirements

The network infrastructure must support both wired and wireless communications.

Wired Connections

Wired connections are assigned to users requiring low latency and high reliability.

User Type	Quantity
Players	50

Administrators 20

Total Wired Users 70

Wireless Connections

Wireless access is provided primarily to spectators.

User Type Quantity

Spectators 130

Total Wireless Users 130



5. Network Topology

The proposed topology follows a hierarchical star architecture.

Core Components

* 1 Core Router

* 1 Core Switch

* 2 Managed Access Switches (48-Port Gigabit)

* 6 Wi-Fi 6 Access Points

* Structured CAT6 Cabling

* Network Monitoring Station

VLAN Segmentation

VLAN ID	Purpose
VLAN 10	Players
VLAN 20	Administrators
VLAN 30	Streaming Team
VLAN 40	Spectators
VLAN 99	Network Management

This design improves network security, performance, and traffic isolation.



6. Power Requirements

Gaming Equipment

50 Gaming PCs $\times$ 500W = 25 kW

Displays

50 Monitors $\times$ 40W = 2 kW

Networking Equipment

Switches, Router, Access Points = 1 kW

Broadcasting and Multimedia Equipment

Estimated = 5 kW

Total Estimated Load

$25 + 2 + 1 + 5 = 33 \text{ kW}$

Safety Margin

20% Reserve Capacity

Required Electrical Capacity:

Approximately 40 kW



7. Cooling Requirements

The gaming systems and audience generate significant heat during tournament operation.

Heat Sources

* 50 Gaming PCs

- * 50 Monitors
- * Broadcasting Equipment
- * Approximately 200 attendees

Cooling Strategy

- * Existing centralized air-conditioning system.
- * Indoor temperature maintained between 20°C and 24°C.
- * Continuous airflow directed toward gaming zones.
- * Temperature monitoring throughout the event.

The cooling infrastructure available in Dewan Canselor is sufficient to maintain operational stability.



8. Security Measures

Physical Security

- * Controlled entrance and exit points.
- * Equipment registration before installation.
- * Security personnel on-site.

Network Security

- * Firewall protection.
- * VLAN isolation.
- * WPA3 wireless authentication.
- * Access control policies.
- * Real-time traffic monitoring.



9. Event Logistics

Setup Phase

Duration: One day before the event

Activities:

- * Equipment installation.
- * Network configuration.
- * Connectivity testing.
- * Performance validation.

Tournament Day

Activities:

- * Participant registration.

- * Match management.
- * Live streaming.
- * Technical support.

Dismantling Phase

Activities:

- * Equipment shutdown.
- * Cable removal.
- * Venue restoration.
- * Final inspection.

10. Conclusion

Dewan Canselor UTeM is the most suitable venue for the proposed Esports Tournament. The facility provides sufficient space, electrical capacity, cooling infrastructure, and network deployment flexibility. The proposed design supports 200 users through a combination of wired and wireless connectivity while ensuring high performance, reliability, and security throughout the event.